# Runway

## Study, video and pitch: understand it, record it, present it

For Sukhman. Use it in this order: read Parts A–C (30 minutes), record the video with Part D, rehearse Part E, drill Part F, then tick Part G on the day.

**Live link:** https://runway-1085585470420.us-central1.run.app

| Part | What it gives you | Time |
|---|---|---|
| A. The problem | The business problem at three depths: 10 seconds, 1 minute, full | 10 min |
| B. What we built | Each piece: the problem it solves, how, the proof, where it is on screen | 15 min |
| C. Numbers to know | The 15 numbers judges will hear; say them the same way every time | 5 min |
| D. The 2-minute video | Second-by-second script and what's on screen | 1 hour to record |
| E. The live pitch | 3-minute and 60-second versions, plus the live twist routine | Rehearse 5 times |
| F. Questions | The hard questions with one-line answers, by judge type | Drill until automatic |
| G. Final checklist | Devpost, demo laptop, the day, after judging | Tick as you go |
| H. Weaknesses and rating | What to admit first, and where we stand | 5 min |

# A. The problem

## In 10 seconds

> Banks judge customers by their **average** balance. Overdrafts happen at the **lowest** balance. Runway finds customers who keep dipping close to $0 before their first overdraft, and tells the bank what to do.

## In 1 minute

Picture someone who earns well and has about $18,000 in the account on average. Every bank dashboard says they're fine. But their pay comes in on the 25th and their bills leave on the 1st, so for a few days each month the balance sits near zero. One surprise (a car repair, a cut in hours, a new loan payment) and they go overdrawn.

Overdrafts matter more than they look. In this data, **every one of the 76 borrowers who was ever overdrawn defaulted on their loan**. So an overdraft is often the first visible sign of a customer heading for real trouble, and of a loss for the bank.

The usual tools miss it because they use averages. The Tangerine track's own runway formula (months of savings left after a shock) uses medians and an average balance. It says many of these customers have “no shortfall” right up to the month they go negative.

## In full: why this is a real business problem

| Who loses | How |
|---|---|
| The customer | Overdraft and NSF fees (now capped at $10 per NSF in Canada), declined payments, stress, damaged credit |
| The bank | Credit losses: overdrawn customers default (76 of 76 here); loans granted to customers already in trouble (27 of 27 defaulted); lost trust and accounts |
| Advisors | No list of who to call, or why; generic $100 balance alerts that fire too late (1 of 34 in our backtest) |
| Regulators | Canada's OSFI Guideline E-23 (from May 2027) expects any model used on customers to be documented, validated and monitored |

## The challenge the track set

Tangerine gave 4,500 customer accounts (1990s Czech bank data, re-dated to 2020–2025) and a runway formula as the answer key. Judges can pick any account and read out a “twist card” (a job loss, rent going up, a repair) and expect an answer live. Our job: say whether the customer makes it, why, and what to do, and find something new in the data.

# B. What we built, piece by piece

Each row is one idea. Learn the left column as the question, the right columns as your answer.

| Problem | What we built | Proof | On screen |
|---|---|---|---|
| Averages hide the danger | **Liquidity trough**: lowest ÷ average balance over 6 months. Flag the bottom 10% | Catches 34 of 55 first overdrafts (62%) in 2025; 13–18× higher risk, three years running, on years the model never saw | 01 headline; 02 trough card |
| The judges' formula misses real life | **Realistic runway**: EI capped at $729/week and ending, pensions kept, the in-month dip, new loan payments counted | Predicts 2025 overdrafts better: AUC 0.84 vs 0.78 (p = 0.002) | 02 both runways; 05 first table |
| A big loan can hide in plain sight | Loan payments count from the month granted | 93172: 120+ months → 2.4 months; overdrawn the next month | 02 story strip, loan card |
| “13.8% chance” must mean 13.8% | Monte Carlo (2,000 futures from the customer's own months), then calibrated | Raw it was 8× too pessimistic; calibrated: 1.61% predicted vs 1.61% actual | 02 forecast card; 05 calibration |
| Different causes need different actions | Diagnosis: money in falling → call; big swings → auto-save; heavy new loan → loan review; dip only → alert + cushion | Money in falling: 20% vs 8% overdrew (2025) | 02 “Why they're at risk”; 04 “Do this first” |
| Advisors need a list, not a dashboard | Whole book: every customer gets one tier and one action; a weekly call list | 10% of customers → 64–68% of next year's overdrafts; 57% need nothing | 01 whole book, call list |
| Credit is risky for these customers | Action ladder, safest first: call → free alert → free transfer from savings → a priced limit last | Alert warned 23 of 34 a median 3 months early; cushion covered 27 of 34; a limit costs ~$108/month at Tangerine's terms | 04 Make a plan |
| Loans go to customers already in trouble | Loan-approval check with the same signal | Holds 17% of loans; they default 34.7% vs 6.2% | 01 lending card; 02 loan card |
| When should the bank act? | Survival analysis (time to first overdraft) | Flagged: 11–12× the hazard, highest in the first 6 months | 02 “When?”; 05 survival chart |
| Why not machine learning? | Champion vs challenger: gradient boosting on 12 features | 39 vs 34, not significant; its top feature is the trough → keep the simple ratio | 05 challenger card |
| Is it fair? Will it keep working? | Fairness by gender and age; 10 monitors with thresholds; model card | No consistent gender gap; 0 breaches | 05 fairness, model health |
| The live twist | AI reads the card into settings (shown as chips); the engine calculates; AI writes the words | Gemini 14 of 16 unseen cards, rules 15 of 16; every number the AI writes is checked | 03 Stress test; 02 assistant |

# C. The numbers to know

| Number | What it means | Say it like this |
|---|---|---|
| 34 of 55 (62%) | First overdrafts caught by flagging 10%, 2025 | “Flag 10%, catch 62%.” |
| 13–18× | Riskiest 10% vs everyone else, each year | “13 to 18 times the risk, three years running.” |
| 76 of 76 | Ever-overdrawn borrowers who defaulted | “Every one of them.” |
| 93172 | Demo customer, Dec 2024 | “$18,500 average, dipped to $238, lent $167K, 2.4 months, overdrawn in January.” |
| 0.84 vs 0.78 | Our runway vs the judges' formula at predicting overdrafts | “Our runway predicts overdrafts better than the formula.” |
| 1.6% vs 1.6% | Forecast vs reality, 2025 | “When we say 1.6%, 1.6% happens.” |
| 23 of 34, 3 months | Personal alert fired early | “Three months' warning.” |
| $0, $0, $108 | Alert, transfer, a limit for a month | “Free steps first; credit last because it costs $108.” |
| 17%; 34.7% vs 6.2% | Loans held at approval; their default rate | “Hold 17% of loans and you hold the ones that default.” |
| 10% → 64–68% | Top tiers' share of next year's overdrafts | “A tenth of the book causes two-thirds of overdrafts.” |
| 57% | Customers who need nothing | “More than half the book needs nothing.” |
| 11–12×, 6 months | Survival hazard ratio; when to act | “Act in the first six months.” |
| 39 vs 34, p = 0.18 | Machine-learning challenger vs the trough | “ML didn't beat one ratio.” |
| 14 of 16 | Gemini on unseen twist cards | “Measured, and checked by chips.” |
| 48 of 57 | Tests that survive multiple-testing correction | “We report the ones that failed too.” |

# D. The 2-minute video

## Setup

- Record the live link or localhost at full screen, browser zoom 110%, Time Machine on **Dec 2024**, account **93172**. Close other tabs and notifications.
- Use Pitch mode: it drives the screens for you. Record the screen first, then add the voice-over (your own voice or ElevenLabs) so you can retake lines.
- Speak slowly: about 140 words a minute. The script below is about 270 words.

## Script with timings

| Time | On screen | Say |
|---|---|---|
| 0:00–0:12 | 01 headline: “Banks watch the average balance. Overdrafts happen at the low point.” | “Banks judge customers by their average balance. But overdrafts don't happen on average. They happen at the low point of the month. And in this data, every borrower who was ever overdrawn defaulted.” |
| 0:12–0:30 | Headline cards, then the whole-book bar | “So we measured the low point. Flag the 10% whose balance keeps dipping toward zero, and you catch 62% of next year's first overdrafts, tested on three years the model never saw. A tenth of customers cause about two-thirds of overdrafts; more than half need nothing.” |
| 0:30–0:58 | Pitch → step 2: customer 93172 story strip, then the loan card | “Take customer 93172. On average, $18,500. The standard formula says no shortfall. But their balance dipped to $238, and that same month the bank lent them $167,000. Counting that payment, their real runway is 2.4 months. They went overdrawn in January. A low-point check at approval would have held the loan.” |
| 0:58–1:20 | Step 3: type the twist card, show the chips, the verdict | “Judges can read any twist card. AI turns the words into settings you can check, the engine does every calculation, and you get a verdict and its biggest reason.” |
| 1:20–1:40 | Step 4: Do this first, then the action cards with $0, $0, $108 | “Then the plan, safest first: call about the cause, a free alert at their own danger level, a free transfer from their own savings. A credit limit comes last, because it costs them about $108.” |
| 1:40–2:00 | Step 5: Why trust it, top cards and the runway-vs-formula table | “Everything is tested on years the model never saw, calibrated, checked for fairness and monitored, and what failed is on screen. Runway: see the low point, explain it, act before the overdraft.” |

## Checklist before you upload

- Numbers in your voice match the screen (93172 at Dec 2024).
- Under 2 minutes; no dead air while pages load (cut it in editing).
- Captions on; the live link in the video description and on Devpost.

# E. The live pitch

## 3-minute version (Pitch mode, press → at each step)

| Step | Time | Say (then stop and let the screen speak) |
|---|---|---|
| Open (before Pitch mode) | 0:00–0:20 | “Raise your hand if you've ever been one surprise bill away from an overdraft. Banks can't see that moment, because they watch averages. We built Runway for Tangerine's advisors to see it coming.” |
| 1 · The finding | 0:20–0:50 | “Flag the 10% of customers whose balance keeps dipping toward zero: we catch 62% of next year's first overdrafts, 13 to 18 times the risk of everyone else, three years running. And overdrafts matter: all 76 borrowers who were ever overdrawn defaulted.” |
| 2 · One customer | 0:50–1:30 | Walk the 4-step strip: “Looks fine, $18,500 average, the formula says no shortfall. The low point: $238. What changed: a $167,000 loan that month, $4,638 a month. Realistic runway: 2.4 months. They went overdrawn in January. The loan card: our check would have held it.” |
| 3 · The twist card | 1:30–2:10 | “Read me any twist.” Type it, point at the chips: “this is what the AI read; nothing runs until I confirm.” Read the verdict and the biggest reason. |
| 4 · The plan | 2:10–2:35 | “Fix the cause first: a call this week. Then the free steps. A limit last, and here's its real cost at Tangerine's terms.” |
| 5 · Why trust it | 2:35–3:00 | “Tested out of time, calibrated, fairness-checked, monitored. Our runway beats the formula. And we list what didn't work. Runway: spot, explain, act.” |

## 60-second version

> “Banks watch the average; overdrafts happen at the low point. Flag 10% of customers by their low point and you catch 62% of next year's first overdrafts. Customer 93172 averaged $18,500 and the formula said no shortfall, but they dipped to $238, got a $167,000 loan that month, and were overdrawn in January. Our realistic runway said 2.4 months, and our check would have held the loan. For each customer we name the cause and the safest action first, and everything is tested on years the model never saw.”

## The live twist routine (any account, any card)

1. Type the account → Open. If it says “opened after…”, switch the Time Machine to Today.
2. 03 Stress test → type the card exactly as the judge reads it → Read card.
3. Point at the chips. If one is flagged (AI and rules disagree), fix it out loud: that's a feature.
4. Read the verdict sentence, then the biggest reason in months.
5. Two runways: “the formula says X; realistically Y, because…”.
6. 04 Make a plan: read “Do this first” and the $0 steps.

**Practise on:** 93172 (the story), 20111 (answer key: job loss 3.4, repair 4.0, combo 3.4 months), 20176, and one young account at Dec 2023 (it explains why there's no forecast).

# F. The questions you'll get

| Who asks | Question | Answer (one sentence, then point at the screen) |
|---|---|---|
| Bankers | Isn't the trough just a low balance? | Yes, that's the point: the average and the formula both miss it; the ratio and the plain low point are tied, and both beat the formula. |
| Bankers | Why offer credit to risky customers? | We don't lead with it: call, free alert, free transfer; a limit only after a call, priced at Tangerine's terms. |
| Bankers | False alarms? | The alert pinged 22% of flagged customers who never overdrew; 1 in 4 alerts was followed by a real overdraft vs a 1.6% base rate. |
| Bankers | Is it fair? | Flag rates are within the four-fifths band in both years; the catch-rate gap flips direction and isn't significant; gender and age are never inputs. |
| AI labs | Where's the AI? | Kept to language on purpose: it reads the twist card and writes the words; a bank can't let a model invent numbers, so every number is checked. |
| AI labs | How accurate is it? | On 16 cards never used for tuning: Gemini 14, rules 15; the chips and the disagreement flag catch the rest. |
| AI labs | Why not ML? | We tested gradient boosting: 39 vs 34, not significant, and its top feature was our ratio. |
| Consultants | What's the business case? | The loan check holds 17% of loans that default 5.6× as often; advisors call 10% of customers instead of guessing. |
| Consultants | How would you roll it out? | Three months in shadow mode, then a controlled test (half get the call), then the CRM. |
| Data platforms | Does it scale? | The core signal is one SQL window query, tested identical to the Python on all 4,500 accounts. |
| Professor | Overfitting? | Features up to December, outcome the next year, three separate years; thresholds fixed beforehand; 48 of 57 tests survive correction. |
| Professor | Is it causal? | No, it's an early-warning signal; the pilot's controlled test is how you'd prove the actions work. |
| Anyone | Biggest weakness? | 1990s Czech data and mostly-cash income, so a job loss assumes all non-pension money stops; that's why the pilot starts in shadow mode. |

# G. Final checklist

## Devpost

- Title, one-line tagline: “See the overdraft coming: the low point, not the average.”
- Live link, GitHub repo, the 2-minute video, 3–4 screenshots (story strip, whole book, verdict, Why trust it).
- Write-up from docs/DEVPOST.md; built with: Python, FastAPI, React, Recharts, statsmodels, scikit-learn, Gemini on Vertex AI, Cloud Run.

## Demo laptop (the night before)

- git pull on main; data folder in place; npm run build; the server starts; open the live link too as a backup.
- Google sign-in for Vertex works (top bar says Gemini); then test once with Wi-Fi off (it falls back to rules).
- Charger, HDMI/USB-C adapter, browser zoom set, notifications off.

## On the day

- Time Machine on Dec 2024, account 93172, Pitch mode ready.
- Rehearse the 3-minute pitch twice and the live twist on 20111 once.

## After judging

- Delete the live site and the cloud project (docs/DEPLOY.md, section 4) so it stops using credit and the data comes off the public link.

# H. Weaknesses and rating

## Say these before a judge does

- The data is 1990s Czech bank data, re-dated; not real Tangerine customers.
- Money in is mostly cash, so we can't see salary; a job loss assumes all non-pension money in stops (the slider shows it).
- It's a signal, not proof that the actions work; the pilot's controlled test is the proof.
- Some reasons held in 2025 but not 2024; we use them to pick actions, not as proven findings.

## Strict rating

| Criterion (weight) | Score | Why |
|---|---|---|
| Discovery (35%) | 8.5 | The low point, customers steering back, hidden new loans, the loan rule; failures disclosed |
| Data use (25%) | 9.5 | All 9 files; leaks caught and tested; out of time on three years; fairness; SQL identical to Python |
| Communication (25%) | 8.5 | Story strip, Pitch mode, folded detail, measured AI accuracy on screen; the rest is rehearsal |
| Impact (15%) | 8.5 | Call list, loan rule, real prices, risk appetite, pilot plan; no “dollars saved” (no loss data) |
| Weighted | ≈ 8.8 / 10 | Likely top 3 in the Tangerine track with a real shot at 1st (a judgement: we can't see other entries) |

> What decides it now is you, not the code: a smooth live twist and a clear 60-second story.